Retail Medium Prototype Scorecard Spreadsheet Review

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Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Roof u-value error (RETM25-Envelope)
- Older vintage SHGC errors (RETM26-Envelope, RETM27-Envelope)
- Plug load density for storage error (RETM64-Equipment)

The following are the highest priority comments:

- The type and size of store needs more justification (RETM8-Notes)
- The number of people is half what it should be (RETM22-Zones)
- No customer entrance infiltration included (RETM33-Envelope)
- No loading dock infiltration included (RETM34-Envelope)
- The total static pressure is too high (RETM40-HVAC System)
- No additional lighting power for displays or merchandise highlighting was included (RETM49-Interior Lights, RETM54-Interior Lights)
- Retail occupancy schedules should be different from Thanksgiving to year end (RETM70-Schedules)
- Lighting is fully on during hours retail is open (RETM73-Schedules)
- Lighting schedule should not ramp up and down for retail (RETM74-Schedules)
- Plug load schedule does not match occupancy (RETM75-Schedules)
- Plug load schedule should not ramp up and down for retail (RETM76-Schedules)

The following comments represent other important issues found:

- A large sales canopy is not common for many retail buildings (RETM58-Ext Lights)
- Plug load equipment power density really depends on retail type (RETM67-Equipment)
- The rooftop efficiency based on a larger size than commonly used (RETM41-HVAC System)
- Common types of zones for retail buildings are missing (RETM17-Zones)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- Why is the medium retail store this size? (RETM3-Notes, RETM4-Notes)
- Why is the window-to-wall ratio 7.1% (RETM14-Prototype)
- Why are the zones the size they are? (RETM21-Zones)
- What is the skylight to roof ratio? (RETM30-Envelope)

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Water Heater
- Interior Lighting
- Ventilation
- Ext Lighting
- Equipment
- Schedules
- Energy Usage
- Weights
-

The comments on these tabs are labeled using the following format RETMx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the Schedules tab.

A copy of the spreadsheet is available that includes which cells on each tab were checked against the sources and were consistent (shown in light green) and which had anomalies and are described in a comment below (shown in tan).

RETM1-Notes (n/a)

A graph that describes the construction rate for retail buildings is missing and is present in other prototype scorecards. This graph is used to justify the divisions and the weighting of different vintages of buildings.

RETM2-Notes (n/a)

Given the complexity of the overall retail market and the wide variety of buildings used in that market, it is surprising that no additional graphs and analysis have been provided.

RETM3-Notes (n/a)

Since there are multiple retail buildings in the set of prototypes, the Notes tab should include justification of the segmentation of the retail building market into the various retail prototypes. Along with this, it should include justification of the floor area and number of floors for the retail market segment the building prototype represents.

The study "Process Evaluation of the PG&E 2006-08 Retail & Hospitality Program" (2011) showed the following table, which may be used to inform how to disaggregate the retail market by floor area:

Table 13: Square Feet of Location

	Hospitality Percent (N=200)	Retail Percent (N=210)	Total Percent (N = 410)
Less than 1,500 sq ft	15%	9%	12%
1,500 - 4,999 sq ft	28%	35%	32%
5,000 - 9,999 sq ft.	13%	15%	14%
10,000 - 24,999 sq ft	5%	15%	10%
25,000 - 49,999 sq ft	5%	12%	9%
50,000 - 74,999 sq ft	2%	1%	1%
75,000 - 99,999 sq ft	2%	1%	1%
Over 100,000 sq ft	7%	3%	5%
Don't know	25%	10%	17%

RETM4-Notes (A8:C8)

The area of the prototype, 24,692 ft2, is not justified except to reference previous prototypes. The Building Types Basics for Retail and Mixed-Use Facilities (2004) includes the following table of typical store sizes:

TYPICAL STORE SIZES FOR MAJOR RETAIL CATEGORIES

Store Type	Size Range (sq ft)
Warehouse clubs	110,000–135,000
General merchandise and discounters	100,000–130,000
Home improvement stores	100,000–130,000
Supercenters	125,000–180,000
Supermarkets and combo stores	55,000–200,000
Sporting goods stores	50,000–60,000
Catalog showrooms	50,000
Toy stores	45,000
White goods stores	35,000–50,000
Furniture stores	35,000–40,000
Baby goods stores	35,000
Home electronics stores	32,000–58,000
Bookstores	25,000–45,000
Apparel stores	25,000–45,000
Super pet stores	20,000–35,000
Computer stores	25,000–45,000
Office supply stores	20,000–45,000
Athletic shoe stores	20,000
Music stores	15,000
Drugstores	8,600–15,000

The "General Merchandise 50% Energy Savings Technical Support Document" (2009) assumes a prototype of 40,500 ft2 for a store.

RETM5-Notes (n/a)

The following sources were used to substantiate one or more comments provided:

- Building Type Basics for Retail and Mixed-Use Facilities (2004).
- Defining the Prototype Convenience Store Building Characteristics – ORNL/TM-2022/2822, December 2022.
- Monitoring and Characterization of Miscellaneous Electrical Loads in a Large Retail Environment. NREL/TP-5500-60668. 2014.
- Final Optimization Report: Empowering Energy Efficiency in Existing Big-Box Retail/Grocery Stores. NREL/TP-5500-77695. 2020.
- Retail Building Guide for Entrance Energy Efficiency Measures, NREL/BK-5500-52290. March 2012.
- High-Efficiency Retrofit Lessons for Retail from a SuperTarget. NREL/CP-5500-56325. 2013.
- Process Evaluation of the PG&E 2006-08 Retail & Hospitality Program. 2011. CALMAC ID PGE0299.01.
- ASHRAE Handbook HVAC Applications, Chapter 2. Retail Facilities. 2023.
- Prototype Commercial Buildings for Energy and Sustainability Assessment: Design Specification, LifeCycle Costing, and Carbon Assessment. 2012. NIST 1732.
- Prototype Commercial Buildings for Energy and Sustainability Assessment: Whole Building Energy Simulation Design. 2011. NIST 1716.
- Analysis and Modeling of Energy Demand of Retail Stores. Building Simulation 2011. Yusuke Suzuki.
- Retail Energy Efficiency Guide. NYC Carbon Challenge.
- Commercial Miscellaneous Electric Loads: Energy Consumption Characterization and Savings Potential in 2008 by Building Type. 2010. TIAX D0498. Kurtis McKenney.
- Technical Support Document: The Development of the Advanced Energy Design Guide for Small Retail Buildings. 2006. PNNL-16031.
- Advanced Energy Design Guide for Medium to Big Box Retail Buildings. 2011. ASHRAE.
- General Merchandise 50% Energy Savings Technical Support Document. 2009. NREL/TP-550-46100.
- Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Nonresidential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual. August 7, 2025. Dimitri Contoyannis.
- California Commercial Saturation Survey. 2014. Itron.
- Effects of Illumination on Store Perception and Shopping Intention: Shedding Lighting on Conflicting Theories by S. Zielke and T. Schienlke (2016)
- Lighting in retail environments: Atmosphere perception in the real world by PJM Custers, YAW de Kort PhD, WA IJsselsteijn, and ME de Kruiff (2010)

- Incorporation of Energy Efficiency into Multi-Tenant Light Commercial Spaces. 2014. PG&E. ET11PGE1121.
- U.S. Lighting Market Characterization Volume I: National Lighting Inventory and Energy Consumption Estimate. 2002. Navigant.
- Retail LED Lighting Assessment. 2009. ET09SDGE0002. Emerging Technologies Associates.
- 2020 U.S. Lighting Market Characterization. April 2024. USDOE. Guidehouse.

RETM6-Notes (n/a)

Some references on other tabs are not clear. On this tab, "PNNL prototypes (90.1-2022)" is used; on other tabs, it is "PNNL Retail_Standalone". On this tab, "CEC prototypes (CBECC 2022.3.1)," but on other tabs simply "CBECC" or "CBECC 2022". On this tab, "DEER prototypes (June 2023)," but on others, "CPUC". On this tab, "2022 Reduced infiltration CASE report," but on other tabs, "2022 CASE report". There are other examples where the citation on the other tabs does not seem to match how the citation is expressed on this tab. This makes understanding the source of the assumptions more difficult.

It would probably be best to have three columns for each cited source. An abbreviation that is used on other tabs, a full citation (APA, Chicago, MLA, etc.), and a URL for how to quickly access the document. Then consistently use the abbreviation throughout the scorecard spreadsheet.

Please confirm that all citations from other tabs are shown on the Notes tab.

RETM7-Notes (C2, C10, C11)

No links are present for three data sources:

- Dodge 2023
- Title 24, part 6, referred to as T24 (1984, 1992, 2001, 2007, 2013, 2022, 2025)
- ASHRAE 90.1 (2022)

RETM8-Notes (A37:B37)

The note describing what the prototype represents is not sufficiently detailed:

- "This prototype represents an open-format "big box" typology for retail up to approximately 100,000 ft^2."

There are many types of retail buildings that still fall into this category, from warehouse clubs, electronics stores, grocery stores, furniture stores, clothing stores, etc. This makes a difference in understanding the prototype since the lighting density, plug load density, use of refrigeration equipment, and other factors vary by the different subtypes of stores.

Other lists of types of retail buildings include Time Saver Details for Store Planning and Design (1995):

- Women's Apparel Stores
- Men's Apparel Stores
- Children's Apparel Stores
- Shoe Stores
- Discount Marts
- Drug Stores
- Food Markets
- Convenience Stores
- Housewares Stores
- Bookstores
- Audio/Video and Computer Stores
- Science and Environmental Stores
- Baked Goods Shops
- Card and Stationery Shops
- Jewelry Stores
- Flower Shops
- Optical Shops
- Museum Shops
- Toy Stores
- Hardware Stores

The list from Building Types Basics for Retail and Mixed-Use Facilities is shown in RETM4-Notes.

A list from the NYC Carbon Challenge:

- Apparel
- Convenience and Pharmacy
- Department Stores
- Financial Services
- Flagship Stores

- Food and Beverage
- Health Clubs
- Wireless Communications
- Specialty Stores
- Supermarkets

The selection of the subtype of store should also be justified based on statistics on this tab of the scorecard. Perhaps to capture the variety of stores,

RETM9-Prototype (A4:B4)

Building type is the first characteristic on the table, but "Medium Retail" is not a building type; it is just the name of a prototype that has been previously used. Perhaps the type of building should be "open-format big box retail" as is stated on the Notes tab. If the intent is really just to represent the middle third of the retail building sizes, some justification should be provided for the distribution of retail buildings by size, as was done for office in the "Geometry of CalBEM Office prototypes_11Nov2025.docx" which used CEUS, CBECS, Dodge, and other data sources to quantify the middle size for offices.

RETM10-Prototype (A6:D6)

Other prototypes have examined the construction trends based on Dodge data for the specific type of building to confirm that the vintages from the general analysis "NonRes-Vintage bins-Draft Report-30Nov2023-Final.docx" were appropriate for the specific prototype building type. This is missing for retail.

RETM11-Prototype (A7:C7, A10:C10, A11:C11)

The rectangular shape, single-story, and flat roof assumptions being made for the retail building prototype seem reasonable and are probably appropriate for a typical retail building based on subjective experience. While we have all seen L-shaped and other-shaped retail buildings, multi-floor retail buildings, and retail buildings with gable roofs and other roof shapes, these seem like reasonable assumptions. It is unfortunate that the only source to justify these assumptions is provided as "PNNL Retail_Standalone, CBECC," pointing to other prototypes rather than statistical sources.

RETM12-Prototype (B7)

The diagram shows that the building windows and glass doors are facing south. While only about 7% of the wall area contains fenestration, this does mean that this building will likely have the highest impact of solar loads of any possible orientation for this building.

RETM13-Prototype (A8:C9, A12:C16)

The values below are only justified using "PNNL Retail_Standalone, CBECC":

Total Floor Area (ft²)	24,692 (178 ft x 139 ft)	PNNL Retail_Standalone, CBECC
Aspect Ratio	1.28	PNNL Retail_Standalone, CBECC
WWR	7.10%	PNNL Retail_Standalone, CBECC
Shading Geometry	none	PNNL Retail_Standalone, CBECC
Floor to floor height (feet)	N/A	PNNL Retail_Standalone, CBECC
Floor to ceiling height (feet)	20	PNNL Retail_Standalone, CBECC
Glazing sill height (feet)	5 ft (top of the window is 8.73 ft high with 3.74 ft high glass)	PNNL Retail_Standalone, CBECC

These values are not intuitively justifiable like a rectangular shape, a flat roof, and a single story. For the remaining values, sources should be cited that are based on statistics, not simply on what previous prototypes have used. Almost all of these figures have probably changed over time, with lot sizes and dimensions changing as well as construction practices. Please find better sources to justify these values.

RETM14-Prototype (A12:C12)

The WWR of 7.10% is very specific and probably has a source that should be tracked down and cited. Another value of 10% is shown in "Prototype Commercial Buildings for Energy and Sustainability Assessment: Design Specification, Life Cycle Costing and Carbon Assessment" (2012) by NIST. This source also assumes a prototype with an area of 8,000 ft2. The "General Merchandise 50% Energy Savings Technical Support Document" (2009) uses a 6.2 WWR with a 22% WWR for the front façade based on reviewing the data in the 2003 CBECS.

I believe from the image that the glazing is all on the front façade. It would also be good to express the WWR for just that façade.

RETM15-Prototype (A9:C9)

The aspect ratio of 1.28 is based on "PNNL Retail_Standalone, CBECC," pointing to other prototypes rather than statistical sources. According to the study "Defining the Prototype Convenience Store Building Characteristics" (2022), the CBECS data set from 1992 provides building shape, length, and width, used to determine a typical aspect ratio. For convenience stores, they chose 1.5 as the aspect ratio. The "General Merchandise 50% Energy Savings Technical Support Document" (2009) uses a 1.25 aspect ratio based on the 2003 CBECS.

RETM16-Prototype (A34:F36)

The four other prototypes in the medium size range, shown at the bottom of the tab in the "background data" section, should be further described. These include the

- CEC: Retail_Medium
- PNNL: Retail_Standalone
- PNNL: Retail_Stripmall
- CPUC: Retail_Grocery

The isometric projections and floor areas, while providing a broad understanding of each building, do not describe the same specific values as the table showing the aspect ratio, WWR, or floor-to-ceiling height. Comparison across multiple prototypes can aid in understanding if the assumptions made for this prototype are reasonable and aligned.

RETM17-Zones (A2:B7)

The retail prototype includes five zones with three different space types:

- Storage (General) - Back_Space
- Retail Sales Area (Retail Merchandise Sales) - Core_Retail
- Retail Sales Area (Retail Merchandise Sales) - Point_of_Sale
- Retail Sales Area (Retail Merchandise Sales) - Front_Retail
- Corridor Area - Front_Entry

All are using the same "Retail" schedule category. But additional space types are probably present in most retail buildings:

- Rest rooms
- Mechanical/electrical/utility rooms
- Offices

- Customer service
- Loading dock
- Break room
- Meeting room

Furthermore, the types of retail space in a typical retail building vary quite a bit, so it may be justified to include these as specific spaces.

RETM18-Zones (A3:C3, A5:C5)

The Point of Sale and Storage zones should have different schedules from other retail spaces. Occupancy of the Point of Sale is higher due to employees who stay there no matter how busy the store is. Likewise, storage occupancy schedules are likely to be longer than when customers are present.

RETM19-Zones (A8)

Three different space types are listed, but this cell says "Total (2 Space Types)".

RETM20-Zones (G2:G7)

The volume calculations are 0.07% too low, which is probably related to the IP to SI conversion factors being used. Multiplying the areas by 20 ft, which is the height of the building, is consistently wrong by a small amount. It might be better to recalculate these numbers in the scorecard rather than include values that were calculated somewhere else.

RETM21-Zones (E3:E7)

What source of information was used to determine the fraction of the building that was storage, point of sale, or entry?

Space Type	Space/Thermal Zone	Area Fraction
Storage (General)	Back_Space	16.6%
Retail Sales Area (Retail Merchandise Sales)	Core_Retail	69.8%
Retail Sales Area (Retail Merchandise Sales)	Point_of_Sale	6.6%
Retail Sales Area (Retail Merchandise Sales)	Front_Retail	6.6%
Corridor Area	Front_Entry	0.5%

This totals 82.9% retail space.

The "Defining the Prototype Convenience Store Building Characteristics" ORNL/TM-2022/2822, December 2022, shows a smaller fraction of retail space, although it is a much smaller building.

Space type	Spaces	Floor area (ft²)	Percentage (%)
Retail	Merchandize, sales, coffee bar/deli/bakery	2,150	71.6
Refrigerated storage	Walk-in cooler	165	5.5
	Walk-in freezer	66	2.2
Office (enclosed)	Office	121	4
Storage	Storage (inventory)	270	9
Utility	Utility	77	2.6
Restrooms	Restrooms	154	5.1
Total	—	**3,003**	**100**

Another source of subdivision of retail space is from NREL's "High-Efficiency Retrofit Lessons for Retail from a SuperTarget" (2013), which is about a larger building including groceries:

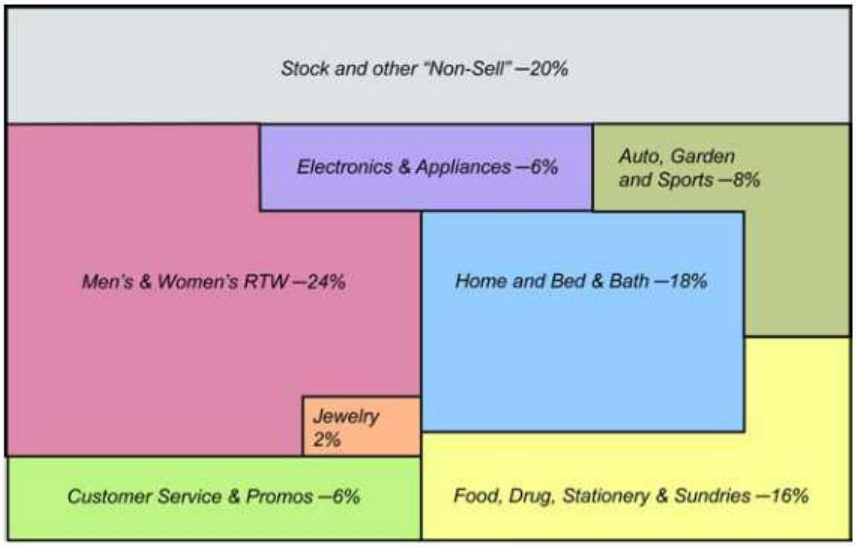
Zone	Area ft² (m²)
General Merchandise 1	722,847 (6,768)
General Merchandise 2	24,960 (2,319)
Grocery	26,520 (2,464)
Stockroom	19,834 (1,843)
Checkout	7,493 (696)
Pharmacy	3,320 (308)
West Vestibule	2,000 (186)
Office	7,700 (715)
Restroom	880 (82)
Food Service	1,840 (171)
East Vestibule	2,000 (186)
Bakery/Deli	4,080 (379)
Total	**173,474 (16,116)**

The "Advanced Energy Design Guide for Medium to Big Box Retail Buildings" contains the table describing the prototype:

Space Type	Medium Box Area (ft^2)
Main sales	31,500
Perimeter sales	3076
Enclosed office	300
Meeting room	500
Break room	500
Restroom	625
Mechanical room	200
Corridor	450
Vestibule	300
Active storage	3050
Total	**40,500**

This is for a very comparable store and shows 85% of the store being sales and 7.5% being storage, but many other specific space types.

This reference also includes the following diagram related to planning for lighting:



Plan schematic for medium box retail footprint

Figure 5-12 Retail Space Planning Schematic

RETM22-Zones (I3:I7)

The formula for the number of people is based on the people/1000 ft2 from the standard and the area, but also includes multiplying by 0.5. This factor that reduces the number of people by half is not explained and is not consistent with other prototypes. It also results in the number of people seeming too low in all spaces.

RETM23-Zones (J3:J7)

The people density "People (Persons/1,000 ft^2) [T24, 2025, Table 120.1-A]" was verified.

RETM24-Zones (n/a)

Given the wide variety of types of retail sales spaces and the corresponding internal loads, it might make sense to create zones for each major type of sales space and weight them to correspond to statistics on how common those spaces are. For example, an area of a store that sells televisions or that sells expensive, detailed merchandise like jewelry consumes a lot more energy than a discount store (think dollar store).

RETM25-Envelope (R8:R9)

The roof u-value requirements for the "pre 1978" and "1978-1997" vintages are shown generally as the same as the values in the "1998-2007" vintage, with the note "Considering 30 years for roof/ insulation replacement." This is consistent except for CZ10, which showed 0.057 for the "1998-2007" vintage value and 0.78 for the "pre 1978" and "1978-1997" vintages. Since the values in other climate zones are 0.78, this is likely a copy-paste error. This is the same comment as NURS36-Envelope and LAB27-Envelope, which implies that the table was copied from one of those spreadsheets.

RETM26-Envelope (I18:X19)

The formula used for calculating the SHGC for "Pre 1978" and "1978-1987" contains an error. The assembly type is described in the cell C18 as "FixedWindow_NonRes." The source is described as "ASHRAE Handbook of Fundamentals for single pane & T24, 2013, for double pane (50%-50%)" but the formula is "=0.8*0.5+I21*0.5" where the 0.8 should be the value taken from the ASHRAE Handbook of Fundamentals, but the table from HOF 2025 shows 0.79 (see page 15.23).

Table 10 Visible Transmittance T_v, Solar Heat Gain Coefficient (SHGC), Solar Transmittance T, Front Reflectance R^f, Back Reflectance R^b, and Layer Absorptance A_n^f for Glazing and Window Systems

ID	Glazing System: Glass Thick., in.	Glazing System: Center Glazing T_v		Center-of-Glazing Properties: Incidence Angles: Normal 0.00	40.00	50.00	60.00	70.00	80.00	Hemis., Diffuse	Total Window SHGC at Normal Incidence: Aluminum: Operable	Aluminum: Fixed	Other Frames: Operable	Other Frames: Fixed	Total Window T_v at Normal Incidence: Aluminum: Operable	Aluminum: Fixed	Other Frames: Operable	Other Frames: Fixed
Uncoated Single Glazing																		
1a	1/8	CLR	0.90	SHGC	0.86	0.84	0.82	0.78	0.67	0.42	0.78	0.78	0.79	0.70	0.76	0.80	0.81	0.72	0.79
				T	0.83	0.82	0.80	0.75	0.64	0.39	0.75								
				R^f	0.08	0.08	0.10	0.14	0.25	0.51	0.14								
				R^b	0.08	0.08	0.10	0.14	0.25	0.51	0.14								
				A_1^f	0.09	0.10	0.10	0.11	0.11	0.11	0.10								

RETM27-Envelope (I19:X19)

The values in rows 18 and 19 should be the same since the note is "ASHRAE Handbook of Fundamentals for single pane & T24, 2013, for double pane (50%-50%)" but the formula for row 19 is "=0.8*0.5+I22*0.5" which referenced T24-2025 row instead of the T24-2013 row. This formula should say "=0.8*0.5+I21*0.5"

RETM28-Envelope (F23:H24, F28:H29)

For the skylight portion of the table, the "Selected T24" for "Pre 1978" and "1978-1987" should say 1992, and the note should reference "143(a) Table NO. 1-I". The table currently shows:

Thermal Properties	Vintage	Selected T24	Note
U-factor IP	Pre 1978	N/A	from 1992 code cycle
	1978-1997	N/A	
	1998-2007	2001	Table 1-H
	2008-2018	2013	Table 140.3-B
	2019+	2025	Table 140.3-B
SHGC	Pre 1978	N/A	From 1992 code cycle
	1978-1997	N/A	
	1998-2007	2001	Table 1-H (For SSR 0-2%)
	2008-2018	2013	There is no requirement in the code (considered same as 2001)
	2019+	2025	

This is simply inconsistent with how the rest of the rows are documented.

RETM29-Envelope (F23:H24, F28:H29)

No justification is provided for why the "Pre 1978" and "1978-1987" values for U-factor and SHGC should come from Title 24-1992. It seems unlikely that skylights would have been retrofitted after 1992. Other rows of envelope values for "Pre 1978" and "1978-1987" are based on ASHRAE, but this one is not. If ASHRAE is not a good source, then perhaps the Pre-1978 value should be the worst U-factor and SHGC values for all climate zones from Title 24 1992, which would be a 1.31 U-factor and 0.86 SHGC.

RETM30-Envelope (A23:X32)

Skylights are specified on the Envelope tab, but no indication of how much skylight area is specified on the Prototype tab. Cell H30 shows "Table 1-H (For SSR 0-2%)", so this may imply that the Skylight-to-Roof-Ratio is between 0% and 2%.

Also note that the "SSR" abbreviation should be "SRR".

RETM31-Envelope (H31:X32)

The SHGC values for skylights for the "2008-2018" vintage and "2019+" vintage have a note, "There is no requirement in the code (considered same as 2001)." The lack of a SHGC requirement was verified, but the assumption of using the 2001 SHGC values seems optimistic. If there is no requirement, it may be assumed to be the worst SHGC value that could be found on the market for each corresponding u-factor, or perhaps just the worst value shown on the SHGC assumptions for any year, which would be 0.86.

RETM32-Envelope (C8:D12, I12:X12)

The roof "Assembly Type" and "Construction" may not be consistent:

Assembly Type	Construction
FlatRoof_WoodFramedAndOther_NonRes	Built-up roof (Metal Decking Roof): Roof membrane+ insulation+metal decking

The "wood framed and other" category and the "metal decking" category (which is assumed to be "metal building") have different criteria in the tables. It seems unlikely that a medium-sized retail store would use a wood-framed roof.

The "2019+" vintage, based on Title 24-2025, uses the "wood framed and other" values (these values are correct), but we presume that a metal built-up roof is what was intended.

The "2008-2018" vintage, based on Title 24-2013, used the "wood framed and other" values (these values are correct), but we presume that the metal built-up roof is what was intended.

For the "Pre 1978" and "1978-1997" vintage that is based on Title 24-1992, the assembly type does not matter.

For the "1998-2007" vintage that is based on Title 24-2001, the assembly type does not matter.

RETM33-Envelope (n/a)

The infiltration related to customers entering and leaving a retail business through the entrance is significant and not described or documented. This level of infiltration is often as large or larger than infiltration through other envelope components. A reference that describes how this can be modeled is: Retail Building Guide for Entrance Energy Efficiency Measures, NREL/BK-5500-52290. March 2012.

Unfortunately, some retail stores are trying to attract foot traffic by simply leaving the door open in the summer, essentially cooling the sidewalk in front of the store, trying to entice people to enter to cool off on a hot day. The effective infiltration for this would be tremendous.

RETM34-Envelope (n/a)

Infiltration from a loading dock may be appropriate to specifically account for in the prototype. Depending on the type of retail, docks may be open for hours per day. Dock seals may or may not be used in a retail environment.

RETM35-Envelope (n/a)

Several envelope table issues were reported for the small hotel that also apply to the hospital prototype table:

- SHSC32 – roof replacement should be weighted
- SHSC38 – consider weighting all standards for historic values

- SHSC39 – shouldn't average single and double pane window criteria since non-physical

For brevity, they will not be repeated.

RETM36-HVAC System (E3)

While the assumption of a PSZAC (furnace) for existing vintages is reasonable, the source cited for CEUS 2022 could not be verified. Please indicate exactly what report and table indicate that this is coming from in CEUS 2022.

RETM37-HVAC System (C14)

The heating efficiency for a packaged heat pump should be expressed in COP at two different operating temperatures, not as EER or SEER.

Vintage	Upgrade/Installed year	Heating	Cooling
Pre 1978	2008	Thermal Efficiency of 80%;	SEER 13 for <65 kBtu/h EER 10.3 for 65-135 kBtu/h EER 9.7 for 135-240 kBtu/h EER 9.5 for 240-760 kBtu/h
1978-1997			
1998-2007			
2008-2018	2013	Thermal Efficiency of 80%;	SEER 13, EER 11.2 for <65 kBtu/h EER 11.2 for 65-135 kBtu/h EER 11 for 135-240 kBtu/h EER 10 for 240-760 kBtu/h
New Construction	2025	SEER 14 for <65 kBtu/h EER 11, COP fed min for 65-135 kBtu/h EER 10.6, COP fed min 135-240 kBtu/h EER 9.5 for >240 kBtu/h	SEER 14 for <65 kBtu/h EER 11, COP fed min for 65-135 kBtu/h EER 10.6, COP fed min 135-240 kBtu/h EER 9.5 for >240 kBtu/h

RETM38-HVAC System (E13)

The reference for the "2008-2018" vintage using T24-2013 shows "2013 Title 20 Appliance Efficiency Regulations, 2013 T24 Tables 112-A, 112-B" but the actual table number is 110.2-A for cooling equipment and 110.2-J for warm air furnaces.

RETM39-HVAC System (C10:C12)

Title 24-2008, which is the basis for the "Pre 1978", "1978-1997", and "1998-2007" vintages, does not appear to have a furnace thermal efficiency requirement in it. I did not check if this was part of Title 20-2005.

RETM40-HVAC System (D19:D23)

The total static pressure assumed: 4.09 in for existing and 3.37 in for new construction, seems much higher than what was used in "Technical Support Document: The development of the Advanced Energy Design Guide for Small Retail Buildings," shown below, or the table for "General Merchandise 50% Energy Savings Technical Support Document" shown after that.

Table 7-3 Baseline Building Calculated Total Fan Static Pressure Drops

Component	Strip Mall Prototype Baseline 5-ton Packaged Rooftop Unit (@2000 cfm)	Standalone Prototype Baseline 15-ton Packaged Rooftop Unit (@5250 cfm)
Internal Static Pressure (Inches Water Column)[1]		
Standard DX Coil	0.15	0.79
Gas Heating Section	0.13	0.51
2-in. Plated Filters[2]	0.15	0.29
Economizer[3]	0.00	0.16
Acoustical Curb	0.04	0.13
Subtotal of internal SP	0.47	1.88
External Static Pressure (Inches Water Column)[4]		
Diffuser	0.10	0.10
Supply Ductwork[5]	0.20	0.28
Return Ductwork[5]	0.05	0.06
Grille	0.03	0.03
Fan Outlet Transition	0.20	0.20
Subtotal	0.58	0.67
10 % Safety Factor	0.06	0.07
Subtotal of external SP	0.64	0.74
Total Static Pressure Drops	**1.11**	**2.61**

Notes:
1. Internal static pressure drops were derived from AAON product catalog for RK Series, last updated on July 1999.
2. Used average difference between the clean and dirty filters.
3. For standalone prototype baseline models, if economizer is not required by the Standard, the total static pressure drops will be 2.45 in. w.c., by deducting the pressure drop of 0.16 in.w.c. from 2.61 in. w.c..
4. External static pressure was calculated based on the typical duct runs served by the listed cooling capacities.
5. Used standard practice of 0.1 inch/100 ft friction rate for the baseline prototypes.

Table 3-21 Baseline Fan System Total Pressure Drops

Component	Package Rooftop, Constant Volume, 10-ton, 4000 cfm, no Economizer (in. w.c.)	Package Rooftop, Constant Volume, 10-ton, 4000 cfm, with Economizer (in. w.c.)
Internal Static Pressure Drop	0.67	0.76
External Static Pressure Drop	0.86	0.86
Total Static Pressure Drop	1.53	1.62

* Used friction rate of 0.1 in. w.c./100 ft (25 Pa/30 m) for the baseline duct pressure drop.

RETM41-HVAC System (n/a)

While the size of the individual rooftop units is not described, based on the size in C20, which says "CAV (>20,000 cfm)" is over 55 tons, but that may not be typical. The report titled "Final Optimization Report: Empowering Energy Efficiency in Existing Big-Box Retail/Grocery Stores" (2020) by NREL of a Walmart has a list of rooftop units:

Table 2: The 27 RTUs Serving the Various Zones within the Selected Walmart Store, and their Associated Cooling/Heating Capacities, Efficiencies, Areas Served, and Zone Assignments within the OpenStudio Model

RTU#	Model	Cooling (Nominal Tons)	Cooling Capacity (MBH) [a]	Heating Capacity (MBH)	EER	Fan Stages	Supply Fan CFM [b]	Supply Fan Operation	Cooling Stages	Area Served	Zone
1	LGB036	10	120	104	12.1	2	3,700	Continuous	2	Kitchen-Deli	1
2	LGB060	5	60	100	10.1	1	2,000	AUTO	1	MechElecRoom	38
3	LGB060	10	120	188	10	1	3,700	AUTO	2	LobbyWaiting	3
4	LGB060	5	60	100	10.1	1	2,000	AUTO	1	CustomService	39
5	LGB060	5	60	100	12.8	1	1,650	Continuous	1	OfficeGeneral	5
6	LGB060	3	36	60	10.5	1	1,500	AUTO	1	VisionCenter	28
7	LGB060	10	120	104	12.1	2	3,700	Continuous	2	McDonald	6
8	LGB060	10	120	104	12.1	2	3,700	Continuous	2	McDonald	7
9	LGB120	3	36	60	14.3	1	1,200	Continuous	1	VisionCenter	9
10	LGB120	20	240	376	10	1	7,000	AUTO	2	RtL RetailSales	10
11	LGB240	20	240	376	10	1	7,000	AUTO	2	RtL RetailSales	11
12	LGB240	20	240	376	10	1	7,000	AUTO	2	RtL RetailSales	12
13	LGB240	5	60	100	10.1	1	2,000	AUTO	1	Gro GrocSales	41
14	LGB240	10	120	188	10	1	3,700	AUTO	2	RtL RetailSales	14
15	LGB240	20	240	376	10	1	7,000	AUTO	2	RtL RetailSales	15
16	LGB240	3	36	60	14.3	1	1,200	Continuous	1	Pharmacy	32
17	LGB240	20	240	376	10	1	7,000	AUTO	2	RtL RetailSales	17
18	LGB240	20	240	376	10	1	7,000	AUTO	2	RtL RetailSales	18
19	SCA036	20	240	376	10	1	7,000	AUTO	2	RtL RetailSales	19
20	SGA036	5	60	100	10.1	1	2,000	AUTO	1	RtL RetailSales	20
21	SGA036	20	240	376	10	1	7,000	AUTO	2	StockRoom	21
22	SGA036	3	36	60	14.3	1	1,200	Continuous	1	StockRoom FF	33
23	SGA036	5	60	100	10.1	1	2,000	AUTO	1	RestRoom	23
24	SGA060	3	36	60	14.3	1	1,200	Continuous	1	Rear Office	24
25	SGA120	5	60	100	10.1	1	2,000	AUTO	1	Rear Office	37
26	SGA120	5	60	100	10.1	1	2,000	AUTO	1	Rear Office	26
27	SGA120	3	36	60	14.3	1	1,200	Continuous	1	Office	5

[a] Thousand Btu (British thermal units) per hour
[b] Cubic feet per minute

The largest is 20 tons with 7000 cfm. This will impact the selected efficiency since smaller sizes have higher efficiency requirements. The model could still use a single zone unit representing a group of smaller units operating together.

RETM42-Thermostat (n/a)

The current setpoints are based on the values from the ACM Appendix 5.4B

Heating Setpoint (F)	70
Heating Setback (F)	60
Cooling Setpoint (F)	75
Cooling Setback (F)	85

Other possible setpoints are 76°F cooling/67°F heating from "Final Optimization Report: Empowering Energy Efficiency in Existing Big-Box Retail/Grocery Stores." (2020 from NLR)

Another source was to use 74°F or 72°F for the cooling setpoint, depending on the dewpoint from "Retail Building Guide for Entrance Energy Efficiency Measures" (NLR):

> Feedback from Retailer Energy Alliance members suggested that such buildings do not frequently include active humidity controls, but that occupants may manually override default thermostat settings if necessary to maintain comfort. To simulate this effect, the building model lowered the thermostat set point from 74°F to 72°F when the dew point temperature exceeded 57°F. The temperature set point was reset to 74°F each day. A dew point of 57°F represented a threshold above which people start to notice discomfort.

The ASHRAE HOF implies that 72°F to 75°F is the cooling setpoint common for big box retailers.

Supporting the current values, the source "Prototype Commercial Buildings for Energy and Sustainability Assessment: Whole Building Energy Simulation Design" (2011 NIST) uses 70°F for the heating setpoint and 75°F for the cooling setpoint with setbacks of 60°F and 86°F, respectively.

The source "Technical Support Document: The Development of the Advanced Energy Design Guide for Small Retail Buildings" (2006, PNNL) uses 70°F and 75°F for the setpoints for heating and cooling, but limits the setback to 5°F for 65°F and 80°F.

RETM43-Thermostat (n/a)

It is not uncommon to have different setpoints for storage, rest rooms, mechanical/ electrical/utility rooms, loading docks, and break rooms.

RETM44-Water Heater (E8, E12)

The "Hot Water Usage Specification (gal/hr per person)" from "[T24, 2022, Appendix 5.4A, sheet: 2022 SpaceFuncData-Input]" shows 0.18 for the Back_Space and Front_Entry spaces. On the Zones tab, those spaces are described as for storage and corridors. For those two space types, the spreadsheet shows 0 gal/hr per person.

RETM45-Water Heater (B19)

No explanation is given for the formula and why 125 gal/hr is used instead of the sum of the actual estimated usage values.

RETM46-Water Heater (D19)

The water heater capacity for other prototypes uses a 0.6 "WH system factor" to determine the capacity, but the formula in the cell shows a 0.85 value is used instead.

RETM47-Water Heater (B28)

No source is provided for the assumption of using 68°F for the ambient temperature.

RETM48-Water Heater (B23)

The cold water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater.

RETM49-Interior Lights (C5, F5)

No additional lighting power is assumed for the retail sales area. More lighting of merchandise usually results in greater sales of the merchandise, so it makes sense for the retailer's bottom line to utilize lighting as much as possible. In T24-2025, the table allows many specific types of additional lighting power allowances, and for many retail buildings, these are very likely to be used.

Building Type/Use	Primary Function Area	Allowed Lighting Power Density for General Lighting (W/ft²)	Additional Lighting Power Qualified Lighting Systems	Additional Lighting Power Additional Allowance [3] (W/ft², unless noted otherwise)
Retail Sales Area	Retail Merchandise Sales	0.95	Decorative	0.35
Retail Sales Area	Retail Merchandise Sales	0.95	Wall Display MH <= 10'6"	9.5 W/ft
Retail Sales Area	Retail Merchandise Sales	0.95	Wall Display MH 10'7"-14'	11.2 W/ft
Retail Sales Area	Retail Merchandise Sales	0.95	Wall Display MH > 14'	12.7 W/ft
Retail Sales Area	Retail Merchandise Sales	0.95	Floor Display & Task MH <= 10'6"	0.45
Retail Sales Area	Retail Merchandise Sales	0.95	Floor Display & Task MH 10'7"- 14'	0.52
Retail Sales Area	Retail Merchandise Sales	0.95	Floor Display & Task MH > 14'	0.60
Retail Sales Area	Retail Merchandise Sales	0.95	Valuable Display Case	0.50
Retail Sales Area	Retail Merchandise Sales	0.95	General Lighting in the enclosed space of ceiling height > 10'	0.10
Retail Sales Area	Fitting Room	0.60	External Illuminated Mirror[5]	40 W/ea
Retail Sales Area	Fitting Room	0.60	Internal Illuminated Mirror[5]	120 W/ea

A bottom-up evaluation based on floor, wall, and display case areas in different types of retail buildings should be performed to create this estimate. In addition, since the building is over 10' tall, the general lighting addition of 0.10 should be assumed for the entire retail space in the prototype. For many retail types, floor and wall displays of merchandise take up a large fraction of the sales area, and it would not be surprising if those additional lighting power numbers add another 0.5 to 0.6 W/ft2 to the overall retail space, especially for apparel, shoe, housewares, jewelry, flower, optical, stationery, and museum stores. These types of displays still occur in discount, drug, convenience, book, toy, and hardware stores, but probably to a lesser extent.

For existing vintages, the Title 24-2016 version table:

PRIMARY FUNCTION AREA	ALLOWED LIGHTING POWER DENSITY (W/ft²)
Retail Merchandise Sales, Wholesale Showroom Areas	1.2 [6 and 7]

Footnotes for this table are listed below.		
FOOTNOTES FOR TABLE 140.6-C: See Section 140.6(c)2 for an explanation of additional lighting power available for specialized task work, ornamental, precision, accent, display, decorative, and white boards and chalk boards, in accordance with the footnotes in this table. The smallest of the added lighting power listed in each footnote below, or the actual design wattage, may be added to the allowed lighting power only when using the Area Category Method of compliance.		
Footnote number	Type of lighting system allowed	Allowed lighting power density. (W/ft^2 of task area unless otherwise noted)
6	Accent, display and feature lighting - luminaires shall be adjustable or directional	0.30 W/ft^2
7	Decorative lighting - primary function shall be decorative and shall be in addition to general illumination.	0.20 W/ft^2

These could easily add another 0.25 W/ft2 for the entire building.

RETM50-Interior Lights (C3:F3)

The cells labeled "Additional LPD for Portable Lighting (W/ft^2)" should probably match the term in Title 24-2025, "Additional Lighting Power Additional Allowance" (even though it is a little repetitive).

RETM51-Interior Lights (E4)

It may make sense to take the Storage lighting power for existing buildings from an earlier version of Title 24 than 2016. Unlike the front of the store, which is very likely to update its lighting system frequently, the back of the house, in a storage room, is likely to update its lighting system rarely, if ever.

RETM52-Interior Lights (A6:G6)

The Front Entry area is considered a "corridor" on the Zones tab, and here the lighting is classified as a "corridor," but in this case, maybe one of the "entry" or even the "retail sales" category would make more sense. Especially since many retail stores have displays of merchandise near the entry.

RETM53-Interior Lights (E4, G4)

The existing vintage lighting power is taken from T24-2015 and is 0.5 W/ft2 based on the note "LPD of 'All other spaces' is considered for Back_Space" but "Commercial and Industrial Storage Areas (conditioned and unconditioned)" primary function area, which is

a 0.6 W/ft2, would probably be better.

RETM54-Interior Lights (E5)

What a reasonable estimate should be for the lighting power density for the existing vintage depends on the questions being asked of the model. The 1.2 W/ft2 for the retail sales area (based on Title 24-2016) with zero additional lighting power may be reasonable, but probably some additional lighting power is justified for most retail spaces (see RETM49-Interior Lights). But various sources indicate that it might still not be as much lighting power as what might be expected in an existing retail store.

- The ASHRAE Handbook-HVAC Applications, Chapter 2, says: "Heat gain from lighting is not uniform throughout the entire area. For example, jewelry and other specialty displays typically have heat gains of 6 to 8 W/ft2 of floor area, whereas the typical sales areas have an average value of 2 to 4 W/ft2. For stockrooms and receiving, marking, toilet, and rest room area, a value of 2 W/ft2 may be used." And from the same source:

Table 2 Approximate Lighting Load for Older Department Stores

Area	W/ft^2
Basement	3 to 5
First floor	4 to 7
Upper floors, women's wear	3 to 5
house furnishings	2 to 3

Table 3 Typical Installed Cooling Capacity and Lighting Levels: Midwestern United States

Type of Space	Area per Unit of Installed Cooling, ft^2/ton	Installed Cooling per Unit of Area, Btu·h/ft^2	Lighting Density of Area, W/ft^2	Annual Lighting Energy Use,[a] kWh/ft^2
Dry retail[b]	367	33	4.0	16.2
Restaurant	136	88	2.0	8.1
Fast food				
Food court tenant area	160	75	3.0	12.2
Food court seating area	147	82	3.0	12.2
Mall common area	282	43	3.0	12.2[c]
Total	264	45	3.6	14.6

Source: Based on 2001 Data—Midwestern United States.
[a]Hours of operating lighting assumes 12 h/day and 6.5 days/week.
[b]Jewelry, high-end lingerie, and some other occupancy lighting levels are typically 6 to 8 W/ft^2 and can range to 11 W/ft^2. Cooling requirements for these spaces are higher.
[c]5.8 kWh/ft^2 for centers that shut off lighting during daylight, assuming 6 h/day and 6.2 days/week.

- The "Prototype Commercial Buildings for Energy and Sustainability Assessment: Design Specification, LifeCycle Costing, and Carbon Assessment," which is based

on the 1999 Commercial Building Energy Consumption Survey, shows a lighting power of 1.5 to 1.9 W/ft2.

- 1.15 to 1.24 W/ft2 from "Final Optimization Report: Empowering Energy Efficiency in Existing Big-Box Retail/Grocery Stores" (2020)
- A value of 1.16 W/ft2 for retail comes from the "California Saturation Survey" (2014) for retail (Table 5-9)
- Based on audits, 1.55 W/ft2 from "Incorporation of Energy Efficiency into Multi-Tenant Light Commercial Spaces" (2014).
- The source "U.S. Lighting Market Characterization Volume I: National Lighting Inventory and Energy Consumption Estimate" (2002) computed average installed power for lighting of 1.7 to 1.9 W/sqft, although this is an older source.
- Careful measurements of a jewelry store in "Retail LED Lighting Assessment" (2009) showed 5.2 W/ft2 made up of both general "focal point" lighting and display case lighting.
- On the other hand, according to "2020 U.S. Lighting Market Characterization" (2024), mercantile buildings had just lighting of 0.6 W/ft2.

Two sources make it clear that the motivation for retailers to add lighting is that it is likely to improve sales:

- Effects of Illumination on Store Perception and Shopping Intention: Shedding Lighting on Conflicting Theories by S. Zielke and T. Schienlke (2016)
- Lighting in retail environments: Atmosphere perception in the real world by PJM Custers, YAW de Kort PhD, WA IJsselsteijn, and ME de Kruiff (2010)
- As "Building Type Basics for Retail and Mixed-Use Facilities" (2004) states: "Retail store lighting is not a space lighting problem; it is a merchandise lighting problem and should be conceived with a solid sales orientation. Lighting and merchandise presentation work in tandem to differentiate products. Merchandise of display is enhanced by compelling compositions and lighting techniques. The techniques by which the designer can add sales appeal to the merchandise with lighting effects, while still staying within the constraints of the project's budget and applicable energy codes, can provide the client with a noticeable competitive edge."

Compounding this urge for more lighting in retail spaces to attempt to drive greater sales of merchandise is that new lighting options may be available that plug into available outlets or are added to the wiring system without a permit. Even those that were installed under a permit, it is unlikely that building code inspectors will be concerned if the building now exceeds the energy code requirements, since they are focused primarily on safety issues.

RETM55-Ventilation (B3:B7, D3:E7)

It is not clear what occupancy category is being used from Title 24-2025 Table 120.1-A for each zone, but presuming Back_Space is being mapped to "Occupiable storage rooms for liquids or gels", Core_Retail and Point_of_Sale and Front_Retail are being mapped to "Sales (except as below)" and Front_Entry is being mapped to "Common corridors" then the appropriate cfm/ft2 is being used. Equation 120.1-F uses 15 cfm/person, so those are also correct.

As described in the RETM22_Zones comment, the 0.5 multiplier on the number of people will cause all the "driven by people" ventilation rates on this tab to be half of what they should be. This matters for the sales areas since doubling those would mean the ventilation rates would increase, since those would be driven by people, not area.

RETM56-Ventilation (F3, K3)

This is the first mention of a restroom, which is apparently part of the Back_Space. It should probably be described on the Zones tab.

RETM57-Ext Lights (E9)

The parking area lighting power allowance of 0.019 Wft2 is consistent with Title 24-2025 Table 140.7-A for general hardscape lighting power allowance for Lighting Zone 2, but it does not include 0.005 W/ft2 from the "Special Security Lighting for Retail Parking and Pedestrian Hardscape" or the 0.018 W/ft for "Security Cameras" described in Table 140.7-B and shown below:

CONTINUED: TABLE 140.7-B ADDITIONAL LIGHTING POWER ALLOWANCE FOR SPECIFIC APPLICATIONS

All area and distance measurements in plan view unless otherwise noted.

PER SITE: WATTAGE ALLOWANCE PER HARDSCAPE AREA (W/ft²). May be used as additional allowance for applicable illuminated hardscape area on the site.

Lighting Application	Lighting Zone 0	Lighting Zone 1	Lighting Zone 2	Lighting Zone 3	Lighting Zone 4
Special Security Lighting for Retail Parking and Pedestrian Hardscape. This additional allowance is for illuminated retail parking and pedestrian hardscape identified as having special security needs. This allowance shall be in addition to the building entrance or exit allowance.	Not applicable	0.004 W/ft²	0.005 W/ft²	0.010 W/ft²	No Allowance

CONTINUED: TABLE 140.7-B ADDITIONAL LIGHTING POWER ALLOWANCE FOR SPECIFIC APPLICATIONS

All area and distance measurements in plan view unless otherwise noted.

PER SITE: WATTAGE ALLOWANCE PER HARDSCAPE AREA (W/ft²). May be used as additional allowance for applicable illuminated hardscape area on the site.

Lighting Application	Lighting Zone 0	Lighting Zone 1	Lighting Zone 2	Lighting Zone 3	Lighting Zone 4
Security Cameras. This additional allowance is for illuminated general hardscape area. This allowance shall apply when a security camera is installed within 2 mounting heights of the general hardscape area and mounted more than 10 feet away from a building.	Not applicable	No Allowance	0.018 W/ft²	0.018 W/ft²	0.018 W/ft²

RETM58-Ext Lights (A13:C13)

The "Sales canopy" of 0.1 ft2 per ft2 of gross building area ends up resulting in about 45% of the total exterior lighting power. While sales canopies of this size are common for auto sales and nurseries, they are not typical for a clothing store or other common big-box retail store types. In addition, B13 on the Prototype tab shows no shading geometry for the building. These are inconsistent and should be rectified.

RETM59-Ext Lights (B10:C10)

Two building entrances for a 25,000 ft2 retail building seem too few. There are likely to be two entrances for customers at the front of the building and a loading dock at the back of the building, as well as at least one other employee entrance. The referenced CASE report suggests an entrance for each 5000 ft2.

RETM60-Ext Lights (B11)

The referenced report for many of the assumptions on this tab is the 2019 Outdoor Lighting CASE Report, which contains the following table. For retail, they applied the value to only 50% of the floor area. Perhaps the same factor should be applied to the prototype.

Table 36: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for Specific Applications – Table 140.7-B

Assumptions for Statewide Estimates - Specific Applications											
		Applied to % of Building S.F. in Category									
Lighting Allowance	Assumptions	Office, LG & SM	Retail	Restaurant	Food (Grocery)	Warehouse, Ref & NR	Hotel	School	College	Other
Hardscape Ornamental Lighting	**0.1 SF per SF of gross building area**	50%	50%	50%	25%		50%	25%	25%	5%

RETM61-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as provided for other prototype reviews.

RETM62-Ext Lights (A7)

No justification is provided for the use of Lighting Zone 2 (LZ2).

RETM63-Equipment (E3:E5)

Two sources for the plug load density values are included, and they are shown in such a way that it is not clear which source applies to which value. The sources are:

1. Plug load and schedules project with CEC, 2025
2. T24 2022 - Appendix 5.4A: Space Use Data-T24N-2022

The 0.4 value for all three space types is from the first source. No values appear to be from the second source.

RETM64-Equipment (B3, F3)

The plug load density for the "Storage (General)" space type is shown as 0.4, but that does not appear in either of the sources shown and appears to be a typo since it is the value for "Retail Sales Area (Retail Merchandise Sales)" from the Plug Load and Schedules Project with CEC, 2025. The note "Matches Retail/All Other; PNNL prototype =~0.7; T24 Appendix 5.4A =0.2" implies that this value is intended to be for "Retail/All Other" but that descriptor is not used in either of the sources cited. The appropriate value for this is probably from "T24 2022 - Appendix 5.4A: Space Use Data-T24N-2022," which is 0.2 W/ft2 for "Storage, Commercial/Industrial (Shipping & Handling)" and "Storage, Commercial/Industrial (Warehouse)."

RETM65-Equipment (E3)

The source "Plug load and schedules project with CEC, 2025" is not an accurate name for the report. It should be cited as "Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Nonresidential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual, August 7, 2025."

RETM66-Equipment (C3:D5)

The gas power density and refrigeration power columns are shown as blank, but should be shown with zero values. The spreadsheet referenced by T24 2022 - Appendix 5.4A: Space Use Data-T24N-2022 shows values of 0.0 for all of these space types.

RETM67-Equipment (B3:B5)

The type of retail store (see RETM8-Notes) can make a large difference in the equipment power density.

In "High-Efficiency Retrofit Lessons for Retail from a SuperTarget" (2013), while not explicitly computed, the average power density for the store was 1.67 W/ft2 based on the following table. However, most of the power draw was in the food service and bakery/deli areas. The general merchandise areas were much lower.

Table 2: Baseline Building Zone and Internal Load Characteristics

Zone	Area ft^2 (m^2)	Volume ft^3 (m^3)	Lights W/ft^2 (W/m^2)	Electric Equipment W/ft^2 (W/m^2)	Gas Equipment W/ft^2 (W/m^2)
General Merchandise 1	722,847	1,651,209	1.42	0.28	-
	(6,768)	(46,757)	(15.28)	(3.05)	
General Merchandise 2	24,960	565,765	1.03	0.03	-
	(2,319)	(16,021)	(11.09)	(0.34)	
Grocery	26,520	601,126	1.13	0.02	-
	(2,464)	(17,022)	(12.16)	(0.19)	
Stockroom	19,834	449,576	1.30	0.25	-
	(1,843)	(12,730)	(13.99)	(2.69)	
Checkout	7,493	169,849	1.42	-	-
	(696)	(4,810)	(15.25)		
Pharmacy	3,320	75,254	0.58	0.48	-
	(308)	(2,131)	(6.24)	(5.18)	
West Vestibule	2,000	45,334	1.24	2.86	-
	(186)	(1,284)	(13.35)	(30.82)	
Office	7,700	174,535	2.43	1.14	-
	(715)	(4,942)	(26.16)	(12.24)	
Restroom	880	19,947	1.38	-	-
	(82)	(565)	(14.85)		
Food Service	1,840	41,707	1.43	58.86	-
	(171)	(1,181)	(15.39)	(633.58)	
East Vestibule	2,000	45,334	1.24	1.35	-
	(186)	(1,284)	(13.35)	(14.57)	
Bakery/Deli	4,080	92,481	0.85	33.40	31.39
	(379)	(2,619)	(9.15)	(359.57)	(337.86)
Total	**173,474**	**3,932,116**	**228,401 W**	**290,231 W**	**128,059 W**
	(16,116)	**(111,344)**			

In "Monitoring and Characterization of Miscellaneous Electrical Loads in a Large Retail Environment" (2014), they carefully monitored individual miscellaneous electrical loads for a single store and found that the most energy-intensive were shown in the table below. Unfortunately, this study never provided an overall equipment power density for the store, even though it did measure 185 different devices.

Table 3–2 Top 10 Most Energy-Intensive CMELs

CMEL	Energy Use
1. Video game display (includes 3 game consoles, 3 TV screens, and display case lights and ad lighting)	29.7 kWh/day
2. Soda vending machine #1	14.2 kWh/day
3. Refrigerator in fast food area	11.8 kWh/day
4. Soda vending machine #2	10.8 kWh/day
5. Soda vending machine #3	8.84 kWh/day
6. Aquarium pump	6.95 kWh/day
7. Freezer in fast food area	6.78 kWh/day
8. Coin wrapping kiosk	6.55 kWh/day
9. Digital photo center kiosk	6.35 kWh/day
10. Charger for floor washer	6.03 kWh/day

Another table from that report shows:

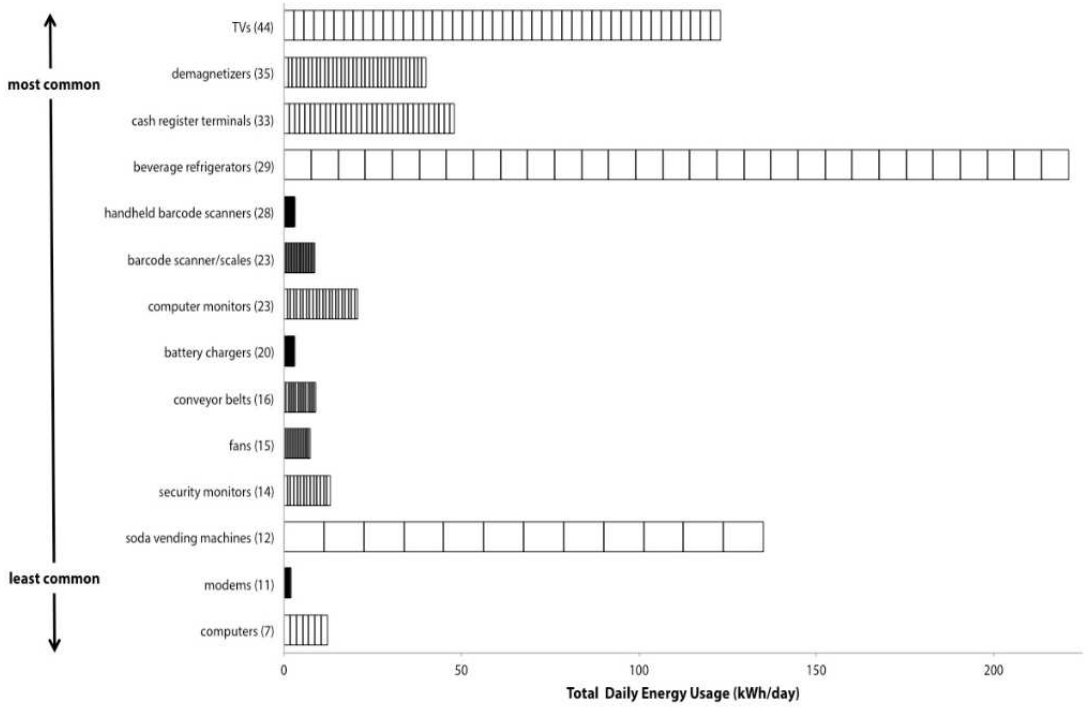


Figure 3–5 The most prevalent devices in the store (quantity is in parentheses)
(The bar represents an extrapolation of the total energy used by devices of that type.)

A Japanese grocery store with measuring equipment power densities had the following table in the paper "ANALYSIS AND MODELING OF ENERGY DEMAND OF RETAIL STORES" (2011). This did not include the refrigeration load. The sales equipment load is equivalent to 0.18 W/ft2. The storage power density is considerably higher at 0.84 W/ft2.

Floor use	Type of heat load	Max heat gain
Entrance	Lighting	33.5W/m^2
	Occupant	0.2 peson/m^2
Sales	Lighting	22.3W/m^2
	Occupant	0.1 person/m^2
	Equipment	2.0W/m^2
Office	Lighting	8.7W/m^2
	Occupant	0.1 person/m^2
	Equipment	2.6W/m^2
Storage	Lighting	6.1W/m^2
	Occupant	0.1 person/m^2
	Equipment	9.2W/m^2
Preparation	Lighting	16.3W/m^2
	Occupant	0.2 person/m^2
	Equipment	64.2W/m^2
Outside	Lighting	10.0W

The "Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Nonresidential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual, August 7, 2025" contains the recommendation to use 0.4 W/ft2 for retail but also includes the following summary table for other sources:

Table 10: Retail Plug Load EPD Top-Down Analysis

Data Source	EPD (W/ft²)
Energy Code NRMFACM	1.00
AEDG - Small Retail (2006)	0.40
AEDG - General Merchandise (2009)	0.20 – 0.70
AEDG - Medium and Big Box Retail (2013)	0.40 – 0.70
ASHRAE 90.1 Appendix C	0.30
ASHRAE 90.1 User Manual	0.25
COMNET	0.55 – 0.82
DOE Reference Building - Retail Stand Alone	0.30
DOE Reference Building - Retail Strip Mall	0.40
ASHRAE 90.1 Prototype - Retail Stand Alone	0.30
ASHRAE 90.1 Prototype - Retail Strip Mall	0.40
ComStock - Retail Stand Alone	0.20
ComStock - Retail Strip Mall	0.40
DEER - Retail Sales and Wholesale Showroom	1.00

Another source, "Commercial Miscellaneous Electric Loads: Energy Consumption Characterization and Savings Potential in 2008 by Building Type" (2010), may contain enough information to derive a value based on individual plug loads, but they do not compute an overall retail sales power draw even though they provide enough numbers to do so.

The source "General Merchandise 50% Energy Savings Technical Support Document" (2009) describes models for both high and low plug load store options:

> "The low plug load level is meant to model stores such as bookstores, which have little to no accent lighting or plug-in merchandise. A typical high plug load store would be an electronics store, which has a much larger amount of plug-in merchandise and accent lighting."

The values used are from the table:

Table 3-6 Peak Plug Loads

Space Type	Low Plug Load Store			High Plug Load Store		
	Electric (W/ft^2)	Accent Lighting (W/ft^2)	Total (W/ft^2)	Electric (W/ft^2)	Accent Lighting (W/ft^2)	Total (W/ft^2)
Main Sales	0.2	0	0.2	0.7	0.8	1.5
Perimeter Sales	0.4	0	0.4	0.6	0	0.6
Enclosed Office	0.75	0	0.75	1.1	0	1.1
Meeting Room	0.75	0	0.75	1.1	0	1.1
Dining Room	2.6	0	2.6	2.6	0	2.6
Restrooms	0.1	0	0.1	0.1	0	0.1
Mechanical Room	0	0	0	0	0	0
Corridor	0	0	0	0	0	0
Vestibule	0	0	0	0	0	0
Active Storage	0.75	0	0.75	0.75	0	0.75
Average			0.29			1.32

The high plug load store shows sales areas of 0.6 to 0.7 W/ft2 and storage of 0.75 W/ft2.

RETM68-Equipment (n/a)

Electric battery forklifts should be assumed for all new warehouses, and a transition to them for existing warehouses should also be assumed based on the typical life of a propane-fueled forklift. This is based on the California Air Resources Board requirement for Zero-Emissions Forklifts:

https://ww2.arb.ca.gov/our-work/programs/zero-emission-forklifts

This impacts internal loads since charging of batteries for the forklifts is a significant load. This is the same comment as WHSC33-Equipment.

RETM69-Schedules (A1:AC30)

The schedule values for occupancy, lighting, electric equipment, HVAC availability, service hot water, infiltration, refrigeration, heating setpoint, and cooling setpoint match their source values.

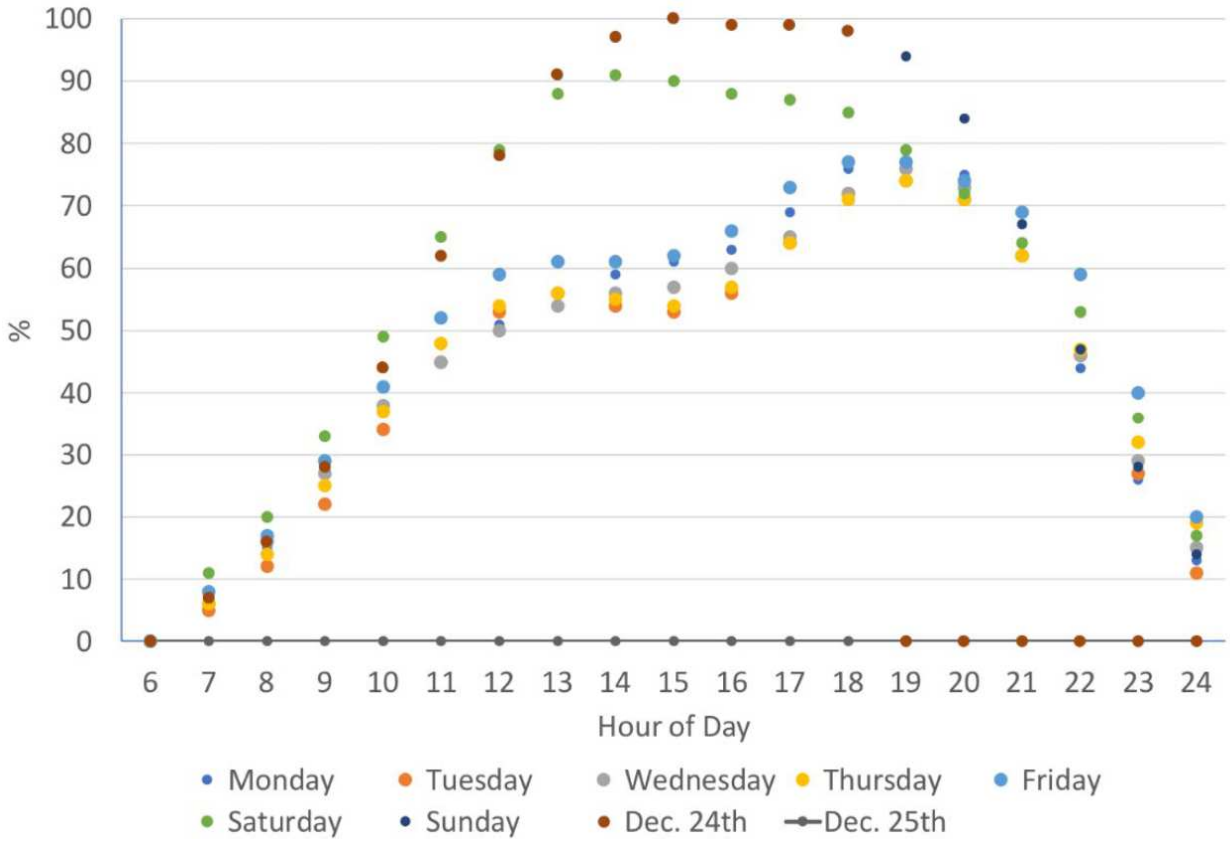
RETM70-Schedules (A4:AC6)

The occupancy level for retail is highly variable throughout the year but generally peaks from Thanksgiving to the end of the year. Based on this, it is unlikely to have a uniform schedule throughout the year. At a minimum, two sets of schedules should be developed, with the much higher levels for the end-of-the-year retail season compared to the rest of the year.

RETM71-Schedules (A4:AC5)

The peak occupancy value is shown as 80% on Saturday from hours 12 to 17, but the peak is also shown on weekdays at hour 16 and at 70% for hours 12 to 15 and 17. This seems unrealistically high for weekday retail. I would expect weekday retail to be half of weekend retail occupancy.

The source "Final Optimization Report: Empowering Energy Efficiency in Existing Big-Box Retail/Grocery Stores" (2020) has a graph of a specific Walmart's occupancy:



RETM72-Schedules (A6:AC6)

Sunday retail occupancy seems too low and is only 40% from hours 13 to 17. Generally, weekends are the best times for retail, and I would expect Sunday occupancy to be near the Saturday retail occupancy level, perhaps 70%.

RETM73-Schedules (A7:AC9)

Unlike other types of buildings, lighting for retail is likely to be fully on, at 100%, during times that the store is open. There are no departments that are not lit or portions of the store that are not fully illuminated during hours the store is open. When skylights provide daylighting, those should be controlled separately without schedule adjustment. The peak lighting schedule value shown is 85%.

To confirm the synchronization of the lighting and occupancy schedules, the source "Incorporation of Energy Efficiency into Multi-Tenant Light Commercial Spaces" (2014) shows that for the small sample, they were very close to the same number of hours.

Sector	Space Type	Number of Sensors	Weekly Lighting Hours	Weekly Business Hours	Lighting /Business Hour Ratio
beauty	salon	1	86.5	73.0	1.18
beauty	restroom	1	25.1	65.5	0.38
common	hallway	1	168.0	168.0	1.00
medical	reception	2	63.6	48.5	1.31
process	laundry	1	156.2	98.0	1.59
restaurant	dining	3	63.9	81.3	0.79
restaurant	kitchen	3	97.1	80.2	1.21
retail	sales	3	80.1	78.3	1.02
service	office	3	112.6	44.0	2.56
Total		**18**	**90.2**	**75.2**	**1.31**

RETM74-Schedules (M7:N8, X7:AA9)

During hours the store is open, there are ramp-up and ramp-down values for the lighting schedules, but that is not appropriate for a retail environment where illumination is provided for all hours the store is open. For example, on Saturdays, the first three hours after the store is occupied are 10%, 30%, and 55%, and the last four hours are 50%, 30%, 30%, and 10%. This is very unlikely for a retail store. At no time would 70% of the lights of a store be off when customers may be present.

RETM75-Schedules (L10, L11, N12)

The plug load schedule is from a different source and is not coordinated for the first hour of occupancy. On weekdays and Saturdays, the occupancy begins at hour 8, but plug loads start at hour 7. This also happens on Sunday.

RETM76-Schedules (L10:M11, Y10:AA11, N12:P12, W12:X12)

The retail plug load schedule is 40% at night and 90% during most occupied hours, but has an intermediate level of 60% for ramp-up and ramp-down (4 or 5 total hours each day). This seems unlikely for a retail environment. These plug loads are TVs, point of service terminals, and other equipment that are on to service or attract customers. It seems unlikely to ramp up and down at the beginning and end of each day. Please explain what equipment might be only partially used when customers are present.

RETM77-Schedules (F16:L18, AA16:AC18)

The service water heating usage in a retail building is primarily driven by restrooms and janitorial sinks, which are both based on building occupancy, but the schedules show nighttime hot water use from 4% to 12% in the early morning and 6% to 27% in the evening when the store is closed. What use of hot water is keeping this from being zero when the building is unoccupied?

RETM78-Schedules (A22:AC24)

The refrigeration schedule is unnecessary since no refrigeration equipment has been described on any other tabs.

RETM79-Schedules (n/a)

The storage "back space" zone is using the same schedules as the sales areas in the store, and that is often appropriate, but for some retail businesses, restocking shelves and displays occurs only during hours without customers, but most retail businesses restock with customers on site. This is a case where the type of retail would influence the kind of schedules needed for a model.